

Mathematical Physics I (Fall 2021): Homework #1 Solution

Due Sep. 17, 2021 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 1, Problem 6.6

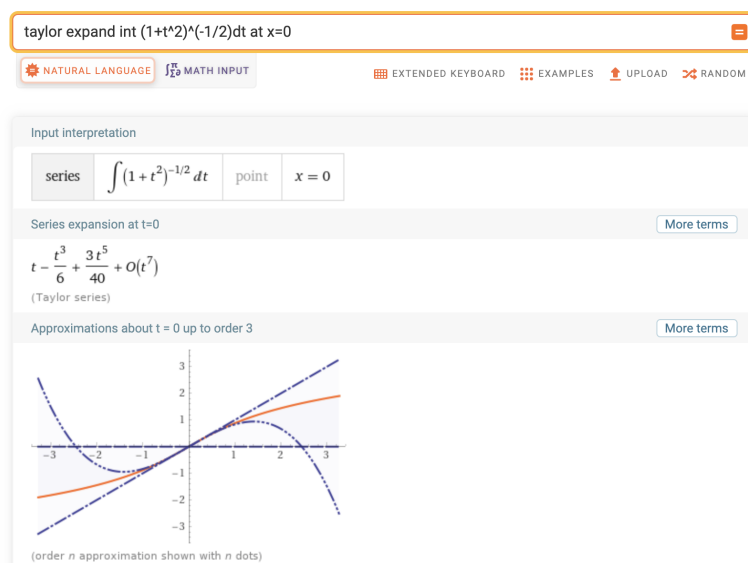
(Note: For Problem 6.6, compare with Oresme's proof demonstrated in Problem 6.2.)

- The series diverges according to the comparison test.

2. Boas Chapter 1, Problem 13.19

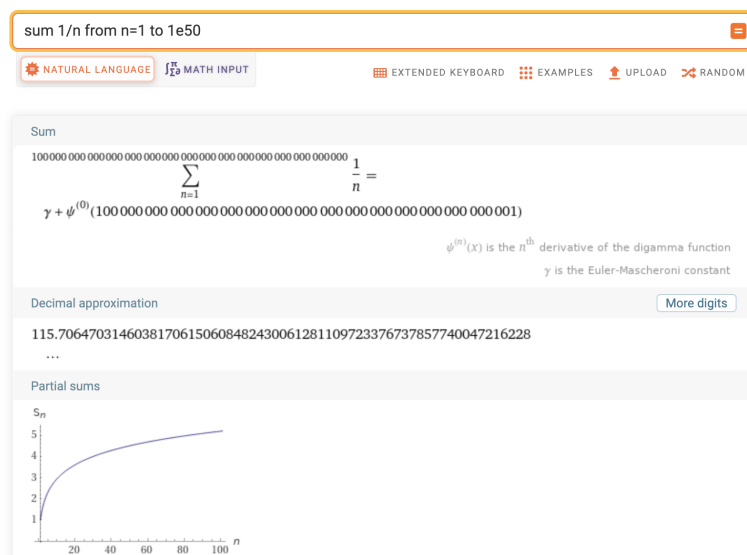
(Note: For Problem 13.19, first prove the equality given, for which you may need to prove and utilize $\sec \theta = \frac{d(\sec \theta + \tan \theta)/d\theta}{\sec \theta + \tan \theta}$.)

- From Eq.(13.5), you can get the binomial series expansion $(1+t^2)^{-\frac{1}{2}} \simeq 1 + (-\frac{1}{2})t^2 + \frac{(-\frac{1}{2})(-\frac{3}{2})}{2!}t^4$. Integrating both sides, you find $\int_0^x (1+t^2)^{-\frac{1}{2}} dt \simeq x - \frac{1}{6}x^3 + \frac{3}{40}x^5$. In a manner similar to Example 13.C2 or Problem 13.4, the general terms are found as $\int_0^x (1+t^2)^{-\frac{1}{2}} dt = \sum_{n=0}^{\infty} \frac{(-1)^n (2n-1)!!}{(2n)!! (2n+1)} x^{2n+1}$.



(Note: For Problem 16.1, it is helpful to draw a diagram that considers the balance between forces and torques, a so-called *extended free-body diagram*.)

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- WolframAlpha® computational intelligence™



- $\rho = \lim_{n \rightarrow \infty} \left| \frac{n(z-i)}{n+1} \right| = |z-i| < 1.$

- Using Eqs.(11.4) and (12.2), $\sin(4 + 3i) = -\frac{i}{2}(e^{-3+4i} - e^{3-4i}) = -\frac{i}{2}(e^{-3}\cos 4 - e^3\cos 4) + \frac{1}{2}(e^{-3}\sin 4 - e^3\sin 4) = i\cos 4 \sinh 3 + \sin 4 \cosh 3$.

(Note: For Problem 15.6, read the instruction in the textbook carefully; that is, you have been asked to compare your result with a computer solution as well. Resolve or explain any disagreement in results.)

- With $u \equiv e^z$, you get $\tanh z = -i = \frac{e^z - e^{-z}}{e^z + e^{-z}} = \frac{u - u^{-1}}{u + u^{-1}} \rightarrow u^2 = \frac{1-i}{1+i} = -i \rightarrow z = \ln u = \ln [\pm(-i)^{\frac{1}{2}}] = \ln [e^{i(\frac{3}{2}\pi + 2n\pi) \cdot \frac{1}{2}}] = i(n\pi + \frac{3}{4}\pi).$

inversehyperbolicatan(-i)

NATURAL LANGUAGE  MATH INPUT

EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Assuming i is the imaginary unit | Use i as a [variable](#) instead

Input

$\tanh^{-1}(-i)$

$\tanh^{-1}(x)$ is the inverse hyperbolic tangent function
i is the imaginary unit

Exact result

$-\frac{i\pi}{4}$

Decimal approximation [More digits](#)

-0.78539816339744830961566084581987572104929234984377645524373614... i

7. Boas Chapter 2, Problem 17.19

- $\arctan z = \omega \rightarrow$ With $u \equiv e^{i\omega}$, you find $z = \tan \omega = \frac{e^{i\omega} - e^{-i\omega}}{i(e^{i\omega} + e^{-i\omega})} = \frac{u - u^{-1}}{i(u + u^{-1})} \rightarrow u^2 = \frac{1+iz}{1-iz} \rightarrow \omega = \frac{1}{i} \ln u = \frac{1}{i} \ln \left(\frac{1+iz}{1-iz} \right)^{\frac{1}{2}} = \arctan z$.

8. Boas Chapter 3, Problem 2.14

(Note: For Problem 2.14, again, read the instruction in the textbook carefully.)

- The equations are inconsistent, so no solution exists.

row reduce {{2,3,-1,2},{1,2,-1,4},{4,7,-3,11}}

NATURAL LANGUAGE  MATH INPUT

EXTENDED KEYBOARD  EXAMPLES  UPLOAD  RANDOM

Input interpretation

row reduce $\begin{pmatrix} 2 & 3 & -1 & 2 \\ 1 & 2 & -1 & 4 \\ 4 & 7 & -3 & 11 \end{pmatrix}$

Result [Step-by-step solution](#)

$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

9. Consider a state of a system (e.g., a particle in a box, or a quantized oscillator) whose energy when occupied by a single particle is ϵ_0 . In the quantum statistical distribution of bosons with unspecified total number of particles (so-called *Bose-Einstein* statistics), the average energy of all the particles in this state is found to be

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} n \epsilon_0 e^{-\frac{n \epsilon_0}{k_B T}}}{\sum_{n=0}^{\infty} e^{-\frac{n \epsilon_0}{k_B T}}}, \quad (1)$$

where k_B is the Boltzmann constant, and T is the temperature. (a) After briefly discussing how this formula is acquired, show that the ratio becomes

$$\bar{\epsilon} = \frac{\epsilon_0}{e^{\frac{\epsilon_0}{k_B T}} - 1}$$

by first identifying Eq.(1)'s denominator as a binomial expansion of $\frac{1}{1-x}$ with $x = e^{-\frac{\epsilon_0}{k_B T}}$, and its numerator as a constant times $\frac{x}{(1-x)^2}$. (b) Using a power series expansion, show that $\bar{\epsilon}$ reduces to $k_B T$ in the classical limit of $k_B T \gg \epsilon_0$.

(Note: You may want to briefly review the classic textbooks in statistical mechanics such as Schroeder or Reif. You are simply asked to come up with a short paragraph about what Eq.(1) means, without laboriously deriving the equation in great detail. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see <http://library.snu.ac.kr/using/proxy>.)

- (a) By differentiating $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, you find $\frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n x^n$, which leads to

$$\bar{\epsilon} = \frac{\sum_{n=1}^{\infty} \epsilon_0 n x^n}{\sum_{n=0}^{\infty} x^n} = \frac{\epsilon_0 x}{1-x} = \frac{\epsilon_0}{x^{-1} - 1}.$$

- (b) With $y = \frac{\epsilon_0}{k_B T} \ll 1$, you get $\bar{\epsilon} = \frac{\epsilon_0}{e^y - 1} \simeq \frac{\epsilon_0}{1+y-1} = k_B T$.

10. By using a complex power series, prove that

$$\sum_{n=0}^{N-1} \cos nx = \frac{\sin \frac{Nx}{2} \cos \frac{(N-1)x}{2}}{\sin \frac{x}{2}} \quad \text{and} \quad \sum_{n=0}^{N-1} \sin nx = \frac{\sin \frac{Nx}{2} \sin \frac{(N-1)x}{2}}{\sin \frac{x}{2}}.$$

Discuss briefly how these series appear in the description of multiple-slit diffraction patterns.

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick.)

- $\sum_{n=0}^{N-1} (e^{ix})^n = \frac{1 - e^{iNx}}{1 - e^{ix}} = \frac{e^{iNx} - 1}{e^{ix} - 1} = \frac{e^{iNx/2} - e^{-iNx/2}}{e^{ix/2} - e^{-ix/2}} = e^{i(N-1)x/2} \cdot \frac{\sin \frac{Nx}{2}}{\sin \frac{x}{2}},$

yielding results that are very similar to one of the Examples in Boas Chapter 2, Section 16.